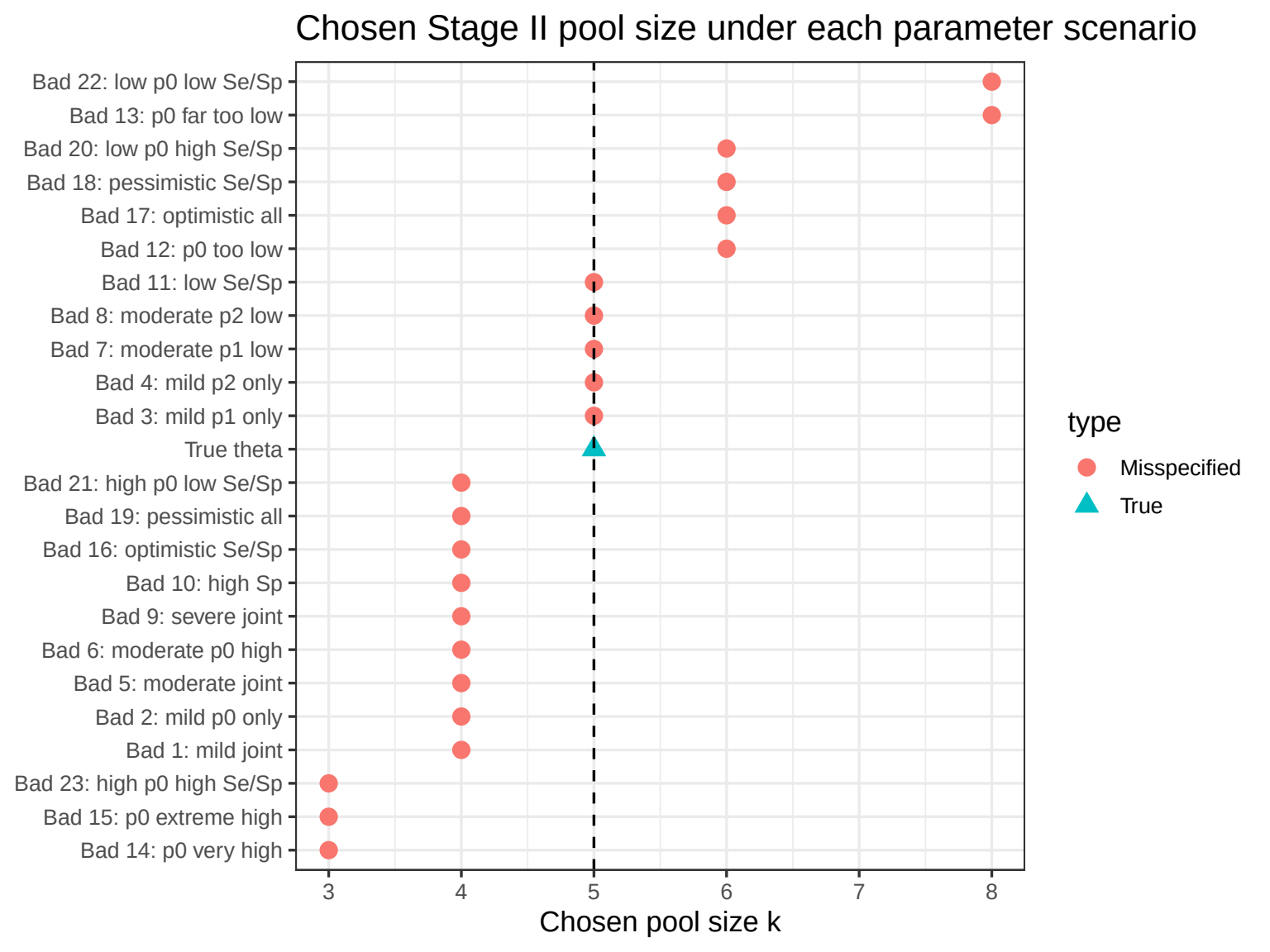
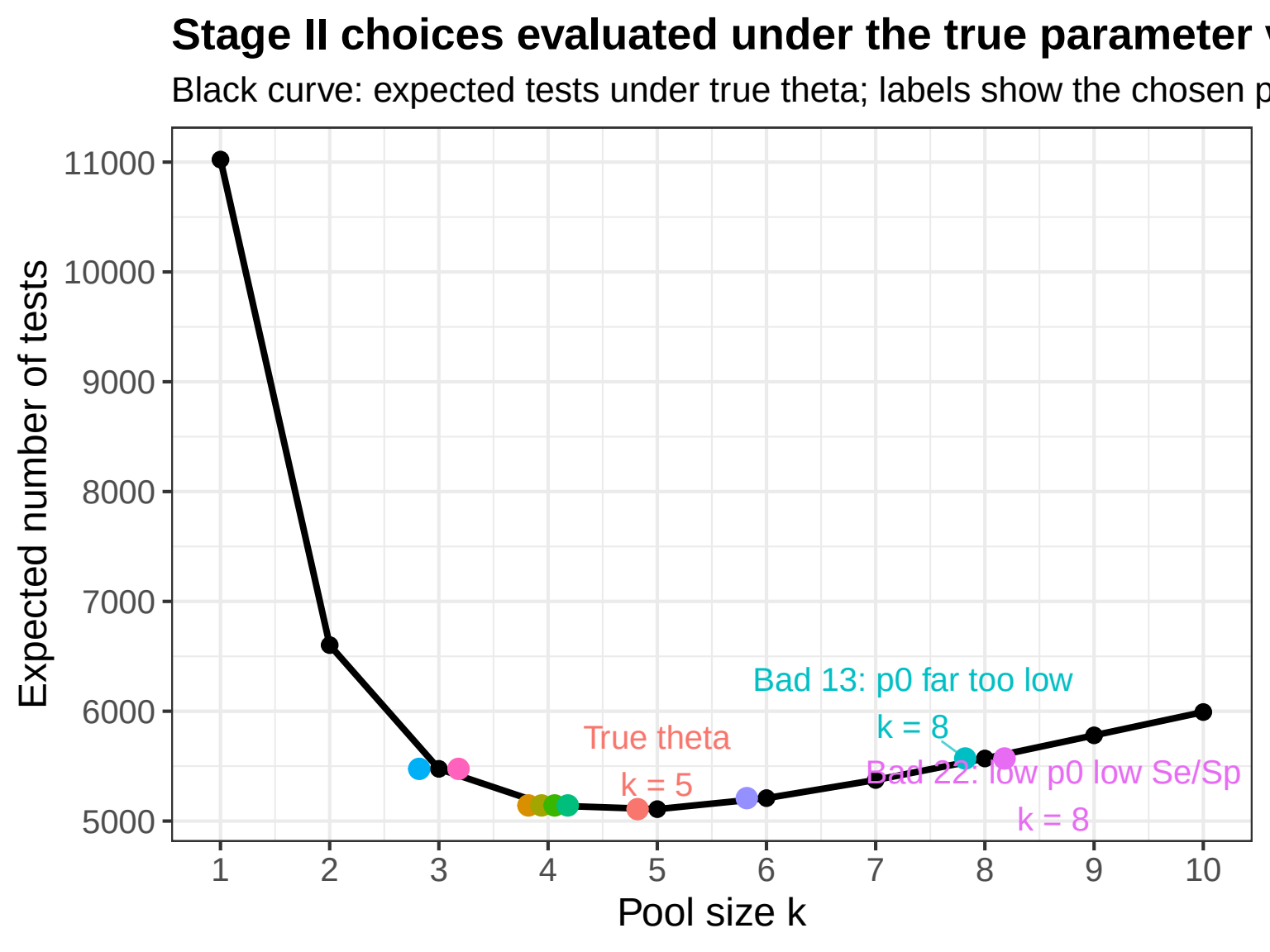
Extra essays needed from using misspecified parameter values
Bad 22: low p0 low Se/Sp
Bad 13: p0 far too low
Bad 23: high p0 high Se/Sp
Bad 15: p0 extreme high
Bad 14: p0 very high
Bad 20: low p0 high Se/Sp
Bad 18: pessimistic Se/Sp
Bad 17: optimistic all
Bad 12: p0 too low
Bad 21: high p0 low Se/Sp
Bad 19: pessimistic all
Bad 16: optimistic Se/Sp
Bad 10: high Sp
Bad 9: severe joint
Bad 6: moderate p0 high
Bad 5: moderate joint
Bad 2: mild p0 only
Bad 1: mild joint
Bad 11: low Se/Sp
Bad 8: moderate p2 low
Bad 7: moderate p1 low
Bad 4: mild p2 only
Bad 3: mild p1 only
True theta
0
100
200
300
400
Extra expected tests under true theta
type
Misspecified
True


Chosen Stage II pool size under each parameter scenario
Bad 22: low p0 low Se/Sp
Bad 13: p0 far too low
Bad 20: low p0 high Se/Sp
Bad 18: pessimistic Se/Sp
Bad 17: optimistic all
Bad 12: p0 too low
Bad 11: low Se/Sp
Bad 8: moderate p2 low
Bad 7: moderate p1 low
Bad 4: mild p2 only
Bad 3: mild p1 only
True theta
Bad 21: high p0 low Se/Sp
Bad 19: pessimistic all
Bad 16: optimistic Se/Sp
Bad 10: high Sp
Bad 9: severe joint
Bad 6: moderate p0 high
Bad 5: moderate joint
Bad 2: mild p0 only
Bad 1: mild joint
Bad 23: high p0 high Se/Sp
Bad 15: p0 extreme high
Bad 14: p0 very high
3
4
5
6
7
8
Chosen pool size k
type
Misspecified
True

Actual expected tests under true theta
Bad 22: low p0 low Se/Sp
Bad 13: p0 far too low
Bad 23: high p0 high Se/Sp
Bad 15: p0 extreme high
Bad 14: p0 very high
Bad 20: low p0 high Se/Sp
Bad 18: pessimistic Se/Sp
Bad 17: optimistic all
Bad 12: p0 too low
Bad 21: high p0 low Se/Sp
Bad 19: pessimistic all
Bad 16: optimistic Se/Sp
Bad 10: high Sp
Bad 9: severe joint
Bad 6: moderate p0 high
Bad 5: moderate joint
Bad 2: mild p0 only
Bad 1: mild joint
Bad 11: low Se/Sp
Bad 8: moderate p2 low
Bad 7: moderate p1 low
Bad 4: mild p2 only
Bad 3: mild p1 only
True theta
5100
5200
5300
5400
5500
Expected tests evaluated at true theta
type
Misspecified
True


Stage II choices evaluated under the true parameter v
Black curve: expected tests under true theta; labels show the chosen p
Expected number of tests
11000
10000
9000
8000
7000
6000
5000
1
2
3
4
5
6
7
8
9
10
Pool size k
Bad 13: p0 far too low
k = 8
True theta
k = 5
Bad 22: low p0 low Se/Sp
k = 8